

CSA1017 – SOFTWARE ENGINEERING

INDIVIDUAL REFLECTION REPORT

INTELLIGENT DISASTER EARLY WARNING AND EVACUATION MANAGEMENT SYSTEM

NAME: MANISHA M

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ROLE: Backend, Database and Alert Management

DESIGN / DEVELOPMENT DECISION:

I contributed to the backend and database design of the disaster-management system. My focus was on organising important system information such as users, disaster alerts, locations, sensor readings, shelters, emergency messages, and alert status. I helped structure the data so that different system modules could access consistent information.

CHALLENGE FACED AND SOLUTION:

The main challenge was managing different types of information generated by multiple system components. I addressed this by organising the data into structured entities and defining relationships between important records. I also considered the need for storing alert information with details such as severity, location, timestamp, and status.

LEARNING GAINED:

I learned how backend systems support the functionality of a complete application. I understood the importance of database organisation, data consistency, relationships, validation, and communication between frontend and backend components.

SDG RELEVANCE:

The project supports SDG 11 by providing an organised platform for managing disaster information. Reliable information management can improve communication and support faster emergency response.

COURSE OUTCOME CONTRIBUTION:

My contribution supports course outcomes related to database design, backend development, software architecture, requirements analysis, and system integration. I learned how properly structured data can support multiple modules within a software system.

POSSIBLE IMPROVEMENT:

I would improve the backend by adding stronger authentication, role-based access control, automatic data backup, improved database optimisation, API security, and scalable cloud-based data storage.